

Task Assignment & Team Roles

Synthea Healthcare Analytics - Power BI, Tableau & Excel Implementation

1. Project Planning & Management Overview

This document establishes the official Work Breakdown Structure (WBS) and delineates roles for the Synthea Healthcare Analytics project. The updated matrix below ensures that all cleaning pipelines, measures creation, and dashboard representations—spanning Power BI, Tableau, and Microsoft Excel—are cleanly mapped to their respective owners for unified presentation and GitHub deployment.

2. Team Responsibility Matrix

Team Member	Data Cleaning Focus	Measures & Logic	Target Dashboards & Tools
1. Asmaa Tarek	Encounters, Conditions	Encounters, Conditions	Power BI: Encounters, Conditions Tableau: Main Dashboard
2. Aya Ayman	Medication, Immunization	Medication, Immunization	Power BI: Medication Analytics, Vaccine Insights Tableau: Dashboard 1 & 2
3. Ghada	Allergies, Careplans + Data Modeling	Allergies, Careplans	Power BI: Allergies, Careplans Excel: Main Reports
4. Alaa Salama	Conditions, Encounters	Observations	Power BI: Main Analytics Excel: Interactive Sheets
5. Habiba	Procedures	Procedures (DAX)	Power BI: Procedures Dashboard
6. Aya Khaled	Patients + SQL Import & Init	Patients	Power BI: Patients Dashboard + Mobile Layout Design

3. Updated Detailed Roles & Responsibilities

[1] Asmaa Tarek

- **Data Cleaning:** Processed and structured the Encounters and Conditions tables.
- **Measures & Logic:** Engineered logical KPIs targeting encounters and clinical conditions distribution.
- **Dashboard & Deliverables:** Created dedicated visual reports in Power BI for Encounters and Conditions, along with a secondary analytical Dashboard in Tableau.

[2] Aya Ayman

- **Data Cleaning:** Cleaned, normalized, and prepped Medication and Immunization data.
- **Measures & Logic:** Calculated metrics for dosage patterns, medication statuses, and vaccination timelines.
- **Dashboard & Deliverables:** Developed Medication Analytics and Vaccine Insights inside Power BI, and completed Dashboard 1 & Dashboard 2 in Tableau.

[3] Ghada

- **Data Cleaning:** Audited and cleaned patient Allergies and healthcare Careplans.
- **Data Modeling:** Successfully structured the central Data Model linking key relational dim/fact tables.
- **Measures & Logic:** Crafted descriptive measures representing active care plans and allergy frequency.
- **Dashboard & Deliverables:** Built Allergies and Careplans dashboards in Power BI, in addition to dedicated Excel analytical views.

[4] Alaa Salama

- **Data Cleaning:** Processed and cleaned the core Conditions and Encounters data.
- **Measures & Logic:** Engineered metrics and calculations for clinical Observations.
- **Dashboard & Deliverables:** Developed an interactive clinical dashboard in Power BI, supported by structured Excel dynamic reports.

[5] Habiba

- **Data Cleaning:** Normalized and formatted clinical and surgical Procedures data.
- **Measures & DAX:** Wrote customized DAX formulas for procedure counts, frequencies, and trend analysis.
- **Dashboard & Deliverables:** Created and designed the Procedures Dashboard inside Power BI.

[6] Aya Khaled (Project Lead & SQL Architect)

- **Database Extraction:** Coordinated initial documentation structure and programmatic extraction of the raw schemas from SQL Server.
- **Data Cleaning & Metrics:** Engineered and cleaned the Patients demographics table, crafting measures for distribution, ages, and patient count.

- **Dashboard & Deliverables:** Developed the primary Patients Demographics Dashboard in Power BI, implementing a cohesive dark visual theme, advanced custom KPIs, and a dedicated, responsive Mobile Layout for executive on-the-go viewing.

4. GitHub Submission & Formatting Guidelines

All components, including local Power BI templates (.pbit), custom Tableau workbooks (.twbx), cleansed Excel files, and documented SQL extraction queries, must be committed to their corresponding directories within the centralized GitHub repository. Clear commit logs and structural clarity must be maintained for seamless evaluation.